

**T.C.
ERCIYES UNIVERSITY
GRADUATE SCHOOL OF NATURAL AND APPLIED SCIENCE
DEPARTMENT OF COMPUTER ENGINEERING**

**DEVELOPING A DORSAL HAND VEIN RECOGNITION
TECHNIQUE BASED ON DEEP LEARNING**

(M. Sc. Thesis)

**Prepared by
Sainaimujiang KUERBAN**

**Supervisor
Asst. Prof. Dr. Mete ÇELİK**

**March 2018
KAYSERİ**

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COMPLIANCE WITH SCIENTIFIC ETHICS

I hereby declare that all information in this document has been obtained and presented in accordance with academic rules and ethical conduct. I also declare that, as required by these rules and conduct, I have fully cited and referenced all material and results that are not original to this work.

Owner of Thesis

Sainaimujiang KUERBAN

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“Developing a Dorsal Hand vein recognition technique based on deep learning” named Erciyes University Graduate Thesis has been prepared accordance with Erciyes University Graduate Thesis Proposal and Thesis Writing Guidelines.

Owner of Thesis

Sainaimujiang KUERBAN

Supervisor

Asst. Prof. Dr. Mete ÇELİK

Head of Department

Prof. Dr. Veysel ASLANTAŞ

ACCEPTANCE AND APPROVAL PAGE

The study called “**Developing a Dorsal Hand vein recognition technique based on deep learning**” has been prepared by Sainaimujiang Kuerban supervised by Asst. Prof. Dr. Mete Çelik, is accepted as a M.Sc. thesis in Erciyes University Graduate School of Natural and Applied Science Department of Computer Engineering by the jury.

...../...../

JURY:

Supervisor : Asst. Prof. Dr. Mete Çelik

Member :

Member :

According to decision dated..... and numbered acceptance of this thesis is approved by Graduate School Administrative Board.

...../...../

Prof. Dr. Mehmet AKKURT

Director of the Institute

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Kayseri, March 2018

DERİN ÖĞRENMEYE DAYALI EL SIRTİ DAMAR TANIMA TEKNİĞİNİN GELİŞTİRİLMESİ

Sainaimujiang KUERBAN

Erciyes Üniversitesi, Fen Bilimleri Enstitüsü
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Tez Danışmanı: Yrd.Doç. Dr. Mete ÇELİK

ÖZET

El sırtı damarı tanıma, biyometrik tanıma tekniklerinden birisidir. Şimdiye kadar çok çeşitli biyometrik yaklaşımlar test edilmiş ancak bazıları istenen performansta sistem doğruluğu verememiştir. Bu nedenle bu tezde, Yakın Kızılötesi (Near Infrared - NIR) görüntüleri kullanarak el damarı kalıplarını tanımlamak ve tanımak için Konvolüsyonel Sinir Ağı (Convolutional Neural Network - CNN) yöntemi ile bir derin öğrenme sınıflandırma stratejisi önerilmiştir. Bu NIR el damarı görüntüleri farklı sıcaklık koşullarına sahiptir ve damar deseninin tekliğiyle daha iyi bir kalite ve ileri güvenlik sağlar. Başlangıçta damar girdisi görüntüleri, yüksek kaliteli el damarı görüntüleri elde etmek için ön işleme tabi tutulur ve el sırtı damarlarının çıkarılması için uygun görüntüler sağlar. Daha sonra, el sırtından damarın parçalanması için Fuzzy C-Mean (FCM) segmentasyon yöntemi kullanılmakta ve böylece özellik çıkarımı adımının desteklenmesi sağlanmaktadır. Damar tanıma evreleri için, dönüşümsel sinir ağı, önceden oluşturulmuş tanımaya göre sınıflandırılmış ilgi bölgesi'ni (region of interest - ROI) eğitir. Ayrıca, CNN yaklaşımı doğruluğu önemli ölçüde artırır ve el damar algılamaya problemine çözüm bulur. Model sonuçlarının analizi görselleştirme teknikleri ile sunulmuştur. Deneysel sonuçlar, önerilen modelin % 94.5 doğruluk elde ettiğini göstermektedir.

Anahtar Kelimeler: Konvolüsyonel Sinir Ağı, Derin Öğrenme, Örüntü tanıma, Sınıflandırma, Biyometrik sistem

DEVELOPING A DORSAL HAND VEIN RECOGNITION TECHNIQUE BASED ON DEEP LEARNING

Sainaimujiang KUERBAN

Erciyes University, Graduate School of Natural and Applied Sciences

M.Sc. Thesis, March 2018

Supervisor: Asst. Prof. Dr. Mete ÇELİK

ABSTRACT

Dorsal hand vein identification is one of the biometric identification techniques. A wide variety of biometric approaches have been tested so far but several propose have been paralyzed the system accuracies. So in this thesis we propose a novel method, deep learning classification strategy by Convolutional Neural Network (CNN) method to identify and recognize hand vein patterns using Near Infrared (NIR) images. This NIR hand vein images have different temperature conditions, and provide a better quality and advance security with its vein pattern uniqueness. Initially, vein input images are pre-processed for high quality hand vein images and which provides suitable images for dorsal hand vein extraction. Then, Fuzzy C-Mean (FCM) segmentation method is used to segment vein from hand dorsal background to support the feature extraction stage. For the vein recognition phases, Convolutional Neural Network is trained Region of Interest (ROI) classified to preformed recognition. Furthermore, CNN approach significantly improves the accuracy and helps us to solve hand vein recognition problem. The analysis of our represented model results is presented with visualization techniques. Experimental results show that our model achieved 94.5% accuracy.

Keywords: Convolutional Neural Network, Deep Learning, Pattern Recognition, Classification, Biometric System.

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
PCA	Principle Component Analysis
SVM	Support Vector Machine
ROI	Region of Interest
RBM	Restricted Boltzmann Machine
DBN	Deep Brief Network
DBM	Deep Boltzmann Machines
DEM	Deep Energy Model
NIR	Near Infrared
AI	Artificial Intelligent
DCNN	Deep Convolutional Neural Network
GUI	Graphical User Interface
BDALab	Brain Diseases Analysis Laboratory
MDC	Minimum Distance Classifier
FCM	Fuzzy C-mean
DL	Deep Learning
RNN	Recurrent Neural Network

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INTRODUCTION

Biometric technology is used for the human feature identification and verification of human body physiological or behavioral characters [1]. Biometric physiological characters includes human facial, fingerprint, palmprint, iris, ear, and vein recognition [2]. On the other hand, biometric human behavioral characters includes gate, signature, and handwriting recognition [2]. During the last decade, vein pattern recognition has attracted researchers in the field of pattern recognition, and has become the research focus for some studies, which include finger vein pattern, hand vein pattern, face vein pattern, palm vein pattern, head vein pattern, heart vein pattern, food vein pattern and iris vein pattern [3]. The most prominent vein pattern recognition types are hand vein pattern recognition and finger vein pattern recognition. This study focuses on the hand vein pattern on the back of the hand.

To date, the application of hand vein recognition technology have been significantly considered in the biometric recognition system for the process of the vein identification and the template for hand images of vein curve due to its stability, uniqueness, and universal personal attributes. Although the hand vein template is widely used, there is lack of comprehensive method to perform hand vein recognition. Moreover, hand vein recognition system often takes in a large amount of data in training before testing. It is similar to other biometric recognition systems as the quality of the images will be affected by the factors such as the environmental illumination, ambient temperature, physiological change, light scattering, user behavior, and inaccurate parameter estimation during the image pre-processing stages of the hand vein recognition system [4].

Despite of the challenges faced by the hand vein recognition system as mentioned above, the improved quality of the hand vein image and the usage of novel classification technology may help to increase the accuracy of the vein recognition. In previous studies, Convolutional Neural Network (CNN) method has achieved a great success in biometric pattern classification. Particularly, the application of the CNN method on

biometric pattern recognition has shown high accuracy and improved over state-of-the-art recognition approaches. CNN is a multi-layered neural network whose perceptions are involved in multiple, overlapping receptive fields that resembles biological visual mechanisms of the visual cortex in most organisms [5].

This thesis focuses on three different fundamental problems. First, methods are designed and established to recover the quality of vein images and vein performance. Second, the Region of Interest (ROI) localization-based algorithm is applied to determine the position of the ROI of hand vein images and CNN algorithm is used to classify hand dorsal vein images. Third, the performances of Principle Component Analysis (PCA), Support Vector Machine (SVM) and CNN are compared and analyzed.

CHAPTER 1

GENERAL INFORMATION AND LITERATURE REVIEW

1.1. Introduction for Biometric System

Nowadays, we are able to access our personal data information from almost everywhere. This is very convenient for information sharing but entails increasing risks as the probability of phishing credentials and data stolen increase with the number of users. More sophisticated protection systems are designed to control possible harassments, such as ID cards cloning, theft or compromised passwords.

Digital protection applications include passwords and smart cards as they are used daily for all digital protection systems. These techniques are being used very frequently. However, passwords and smart card techniques are relatively insecure methods of protection and easier access to the security systems.

The experimental research on biometric systems is growing rapidly in academic and commercial levels. Thus, it is challenging to innovate new techniques and improve the protection of personal information of users.

Biometric recognition systems are applied to which analyze the individual traits such as finger veins, hand veins, palmprint, face recognition, voice recognition, fingerprint, and iris recognition. Biometric recognition systems are widely used to control the access to various applications, private property or even in forensic scenarios. The advancement of the biometric recognition system is available to provide a real time automatic resolution, which extracts human features from individuals, then computes a template and compares it with the one which are stored and saved previously in a database to provide a matching decision.

Deep learning (DL) provides powerful enhanced performance in machine learning tasks. DL methods also have helped to solve the existing problems with the traditional biometric pattern recognition. Hence, this thesis focuses on the deep learning methods

to represent pattern recognition structures in biometric recognition system. In this chapter, it is aimed to review the state-of-the-art in deep learning methods in biometric pattern recognition system by identifying the challenges from recent research articles. First, the overview of various deep learning methods and their design characteristics are summarized. Second, a brief description of their application in biometric pattern recognition in biometric systems are presented, such as face recognition, fingerprint recognition, palmprint recognition, vein pattern recognition, Irish and eyes recognition, and gate recognition.

Many human physical characterizes have been proposed on biometrical system and compared with the other biometric physical characteristics. Vein pattern recognition is one of the latest methods and is considered as a trustworthy biometric technique in current research area. It was developed to improve various issues faced by traditional biometric systems.

Here are the summary and explanation of the reasons to adopt vein pattern on biometric recognition system.

- The acquisition process of vein image does not need direct contact with the vein pattern-extracting sensor [6]. The Vein pattern image is relatively hygienic compare to other pattern trades, thus user identification process will be improved by the vein imaging technology.
- Vein pattern is one of the unique structures to each individual, which doesn't change over the time [7]. Advances in vein vessels in human body provide a unique and stable operational structure for biometric recognition system.
- Vein recognition technology is notably less costly compare to many other biometric technologies (like, iris scanning technology) [6].
- The vein patterns are difficult for intruders to copy, steal or misuse compared to other biometric features [8]. Vein vessels are hidden underneath the human skins and invisible for human normal visions. This advantage makes vein patterns to be one of the most trustable and stable individual features on human body.

1.2. Biometric Recognition System

Biometric recognition, or simply biometrics, refers to the automated recognition of individuals based on their biological and behavioral characteristics [9]. A lot of biometrical traits and features successfully applied to provide security measurement for the modern society. Every biometrical trait represents an individual, which has become very important to identify or authenticate a user application system such as the personal security system in companies, finance area, airport security system, health care system in hospitals, and other important and special areas.

However, personal security system not abandon password or token process from security system when apply biometric applications. While biometrics is applied, it can decrease some kinds of password limitations on user, who associated. Biometric systems have provided vulnerability to spoof attacks, likability attacks in the system usability and system dataset, which reduce the financial burden to provide the user better services.

The main objective of a biometric system is to recognize individual trades with high accuracy. Since biometric traits are generally inherent to an individual, there is strong permanent link between a person and his or her biometric traits [10]. This emphasizes the importance of the development of a customized biometric recognition system which has a low error rates. System error rates refer to metrics of false match rate (FMR) [11] and false non match rate (FNMR) [11] errors in biometric verification system. The identification system includes to false positive identification rate (FPIR) and false negative identification rate (FNIR).

A typical biometric recognition system includes five special models as specified below. Figure 1.1 illustrates the composition of a general biometric recognition system.

- i. Acquisition image model is acquiring individual image from source to system.
- ii. Feature extraction model is processing the salient scanned biometric data to extract information with set of features that are useful in distinguishing between different users [9].

- iii. Pattern matching model is extracting features from the input image and obtaining to match gallery image score.
- iv. Decision model is used to verify or to reject the result on match score in system.
- v. Database model is represented digital container for the acquired samples with templates, which presented users information data.

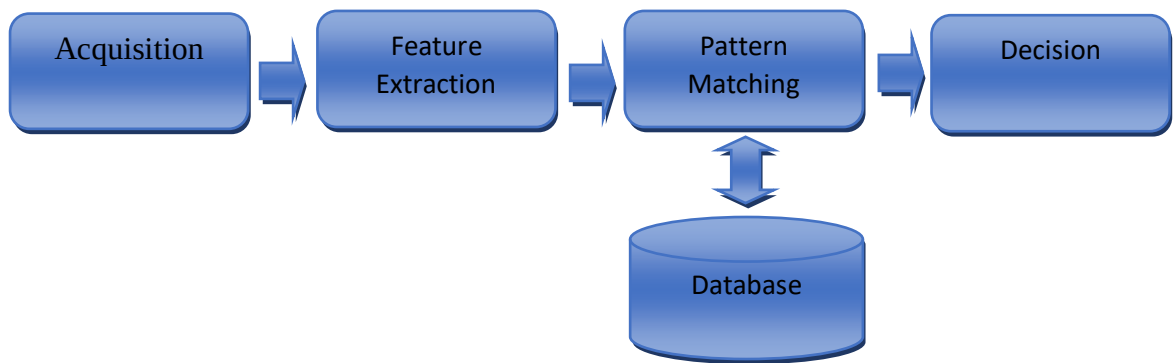


Figure 1.1. Typical Biometric System Model

1.2.1. Biometric Characters

Biometric recognition in individuals represents human biological and behavioral characteristics. Human biological characters include human morphological (physical) characters and human biological characters. Following are some human physical characters:

- Facial pattern recognition is one of the most popular and most commonly used biometric recognition system applied on facial features.
- Fingerprint pattern recognition is applied on human fingerprint for human identification for many decades. Finger ridges and valleys are displayed on personal finger surfaces, which applied to differentiate from each other.
- Iris pattern recognition is a complex process. The iris is the annular iris region of the eyes bounded by the pupil and sclera on either side [12]. People with different eye colors such as brown, blue, green, black and gray colors. Thus, iris patterns also refer complex patterns in human individual trades.
- Palmprint pattern recognition used on human palm and patterns are similar to the human fingerprint ridges, and valleys. It also applied to other special features

such as principle lines and wrinkles with unique patterns. Palmprint images can be collected easily with low cost using the scan resolution devices under normal condition.

- Retinal pattern recognition is applied widely in military propose as it has the most advanced pattern stability compare to others. The retina is a layer at the back of the eyeball that contains cells sensitive to light, which trigger nerve impulses that pass via the optic nerve to the brain, where a visual image is formed. The retinal vascular pattern is characterized using retinal recognition, which can be imaged by exposing the human eye to infra-red light [11].

Human DNA trade is one of the most utilized biological trades in the field of medicine and police forensics. Human behavioral characters include signature, gait, voice and keystroke.

- Signature pattern recognition: Signature is an individual behavior of a person who sign their names with a unique style of their own. Signatures of people are significantly different from each other [9]. The recognition trades are consisted of signature shapes, position of the pen using style, writing pressure, length of strokes, and curvature radiuses measurement.
- Voice pattern recognition: Voice is an individual trade in recognition system used to identify a person. Voice of a person depends on the vocal tract, mouth, nasal activities and lips movement that are used in synthesis of sound [13].
- Keystroke pattern recognition: Keyboard using patterns represent as a behavioral characteristic when people type a computer or any kind of devices.
- Gait pattern recognition: Gait refers to a person's walking style on the way which treated as a behavioral characteristic and can be used to propose an individuals' walking distances. Most of the time, surveillance system us used to identity a person.

1.2.2. Biometric Applications

In the recent years, there is an increasing trend in mass deployment of biometric technologies in human identification in both public and private sectors. Most of the techniques require close contact of the subject with the imaging sensor in order to

establish the identity. However, face recognition and recognition based on gait have the potential to identify humans from certain distance. As a result, face recognition and gait recognition systems are suitable for the video surveillance activities such as the suspect identification at airports, crowd monitoring and human activity recognition at public places which are not possible with other biometric techniques such as fingerprint, iris recognition that require dedicated arrangements for imaging as well as sufficient user cooperation.

- Biometrics is used by the Government Law enforcement. Such systems refer to the biometric techniques that can support government law enforcement agencies. Specially, fingerprint identification and palmprint identification systems are used to process store, search, and retrieve images and records of individual fingerprint or palmprint.
- Biometrics systems are used in border control operations. Such systems are most widely used in European countries to control and monitor border area, which particularly consists of electronic passport or e-passport system. Those systems include a stored database which comprises data of two fingerprints and a passport photo.
- Biometric systems are applied in healthcare and subsidies. Those systems are used in European, Middle Eastern and some African countries to process their ID cards and health Insurance programs. For instance, palmprint identification system is utilized for healthcare insurance program in Turkey, where the system collects palmprint images from specific devices and addition to the ID cards provided by citizens in order to access the healthcare program.

1.2.3. Vein Patterns Apply on Biometric System

Several types of techniques have been used to identify the uniqueness of individuals in biometric system [3]. Vein patterns are the one of those unique individual's characteristics. Vein patterns include finger veins pattern, hand veins pattern, face veins pattern, palm veins pattern, palate veins pattern and food veins patterns.

Nowadays, recognition techniques based on vein patterns are commonly used in biometric security system, as those techniques are not easily modified and falsified. Veins in the human body are stable and unique, so this is an advantage for a biometric system to be reliable and common used in biometric recognition system.

1.3. Deep Learning

Since 2006, deep learning methods are started to be applied in the areas of machine learning and artificial intelligences. Deep learning strategies has great potential to leverage very large datasets for finding hidden structure and also to make accurate predictions[14]. A of machine learning techniques can exploit many layers of non-linear information processing for supervised or unsupervised feature extraction and transformation, and for pattern analysis and classification [15]. In hierarchical feature learning, the goal of deep learning is to extract multiple layers of non-linear features. In this thesis, it is interested in stacking such deep hierarchies of non-linear features because it is not easy to learn complex features from a few layers. A deep learning architecture is a multilayer stack of simple modules, most of which are subject to learning, and many of which compute non-linear input–output mappings [16].

Figure 1.2 presents deep learning application on computer vision architecture and three major classes of deep learning. The major division of deep learning depends on how the architectures and techniques are intended to use for recognition or classification.

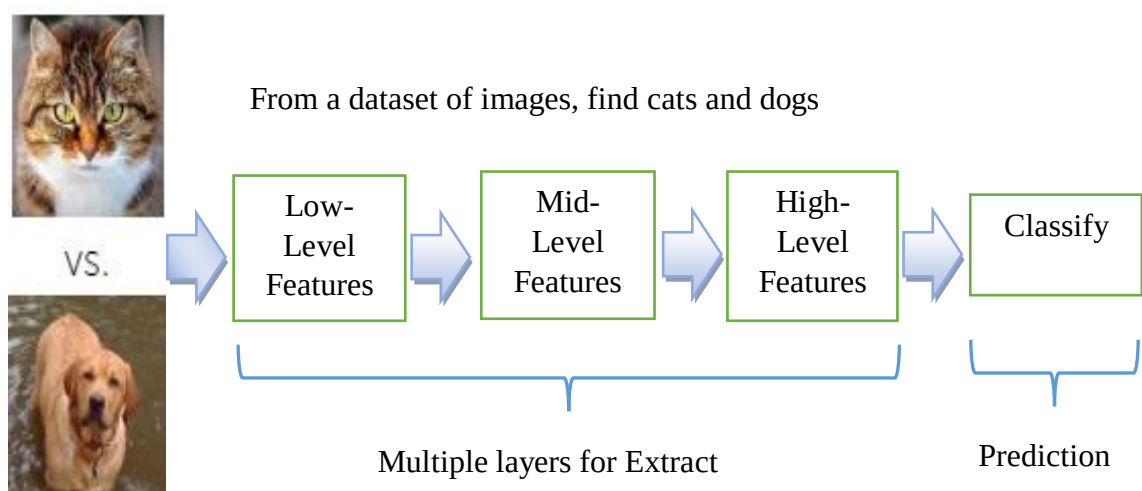


Figure.1.2. Deep Learning Work Flowchart

1.3.1. Convolutional Neural Network

The Convolutional Neural Networks (CNN) is one of the most notable deep learning approach, which consists of a convolutional layer and a pooling layer [15]. It has been found highly effective and is also the most commonly used in diverse computer vision applications. CNN is extended from artificial neural network, the convolutional layer shared many weights, and a pooling layer subsample the output of convolutional layer also reduces the data from the previous layer.

The pipeline of the general CNN applied on computer vision architecture is shown in Figure 1.3. Generally, a CNN consists of three main neural layers, namely, convolutional layers, pooling layers, and fully connected layers. Different kinds of layers play different roles. In Figure 1.3 image classification proposed by a general CNN is shown layer-by-layer. There are two stages for training the network: a feature extraction stage and a classification stage.

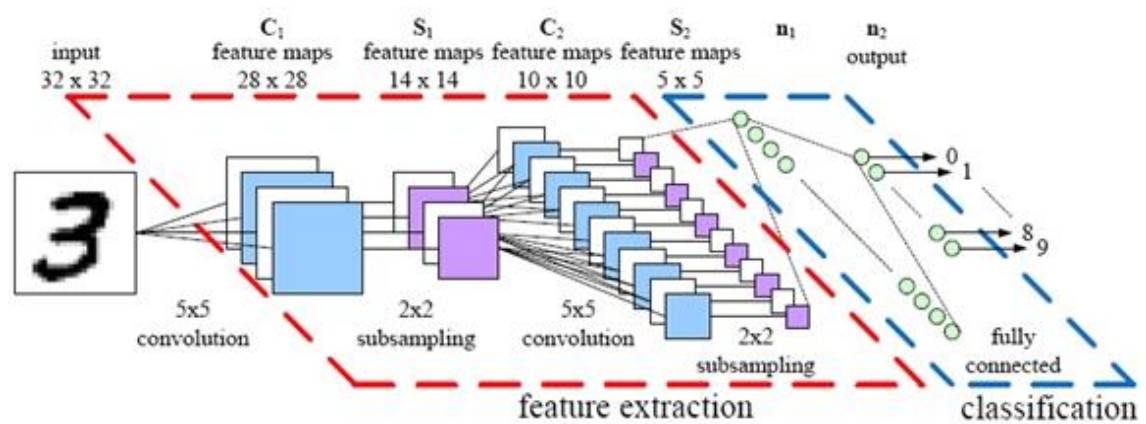


Figure 1.3. Convolutional Neural Network Architecture [46]

1.3.2. Restricted Boltzmann Machines (RBMs)

Restricted Boltzmann Machine (RBM) is using two-layer (hidden neurons) network to consider training set of binary vectors in image files, which consider as an unsupervised learning method. RBM have been used in different type of biometric systems, which is usually trained using the contrastive divergence learning procedure [17]. As can be seen in Figure 1.4 RBM working model includes two-layer includes such as hidden layer and visible layer. Every layer neurons not connected with same layer neurons only

connected with different layer same column and not same column neurons. This advancement of RBM is increasing learning capability.

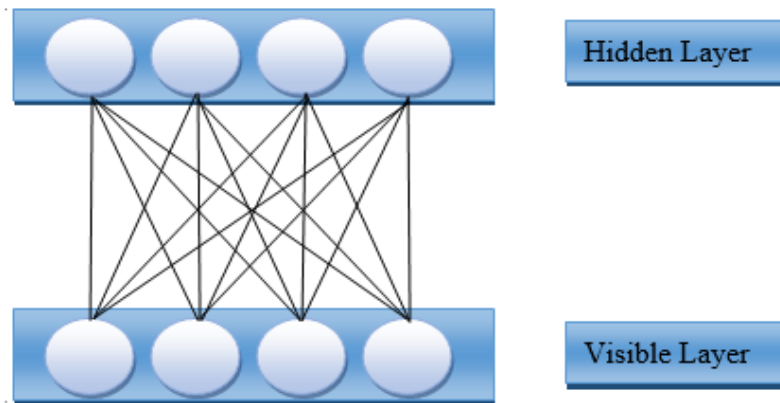


Figure 1.4. Restricted Boltzmann Machine Working Architecture Model [19]

There is an increasing trend in the application of RBM method on biometric recognition systems, such as, face recognition, speech recognition, and fingerprint recognition systems. Face pattern recognition is the most challenging application in image recognition tasks. Early experiment with deep learning method RBM on face pattern recognition has started since 2012. A study by Huang, Lee, and Miller [18] has used convolutional deep belief network and convolutional restricted Boltzmann machines method to extension local CRBM to learn hierarchical representation and training LFW-a face images dataset. In face spoofing detection problem, G. Botelho. Suza et al. have used RBM to train probably data features and reduce the dimensionality of the feature vectors extracted from faces to optimize the dimensionality reduction [19]. In another study [20], researchers proposed a novel method for noisy regions over patches of the fingerprint recognition with RBM, where methods are trained with clean fingerprint images and applied to overlapping patches of the input fingerprint. RBM was applied on speech for text independent speaker recognition to learn lower dimensional feature vectors extracted to get higher recognition accuracy [21].

1.3.3. Autoencoder

Autoencoder is used to obtain a representation of the original data with a reduced dimensionality, which is a neural network that maps $R_n \rightarrow R_n$ with a single hidden layer

of Rectified Linear Unit (ReLU) activation [22][23]. Autoencoder can be divided into two types, namely, undercomplete and overcomplete autoencoders [24]. Figure 1.5 shows a simple architecture of autoencoder method and this method includes three units. First unit is input layer X , second unit is three fully connected hidden layers and third unit is output layer X^1 . Also, the number of input neurons is equal to output neurons. The most advanced autoencoder has two main parts in process operation, which are encoder and decoder. Encoder is a single or a group of layers that allow to reduce the dimension of the input original data, and decoder is performance of the opposite operation to produce the approximate reconstruction of original input data [22]. The research area of deep learning methods on biometric recognition system is gaining more attention recently. The impact of autoencoder method on biometric performance varies significantly. In a sparse autoencoder based on gradient descent has been proposed for finger veins feature extraction [25], for the verification or identification system choosing the best set of discriminative features extracting from input data is one of most challenging processes, which will impact the accuracy of the classification system.

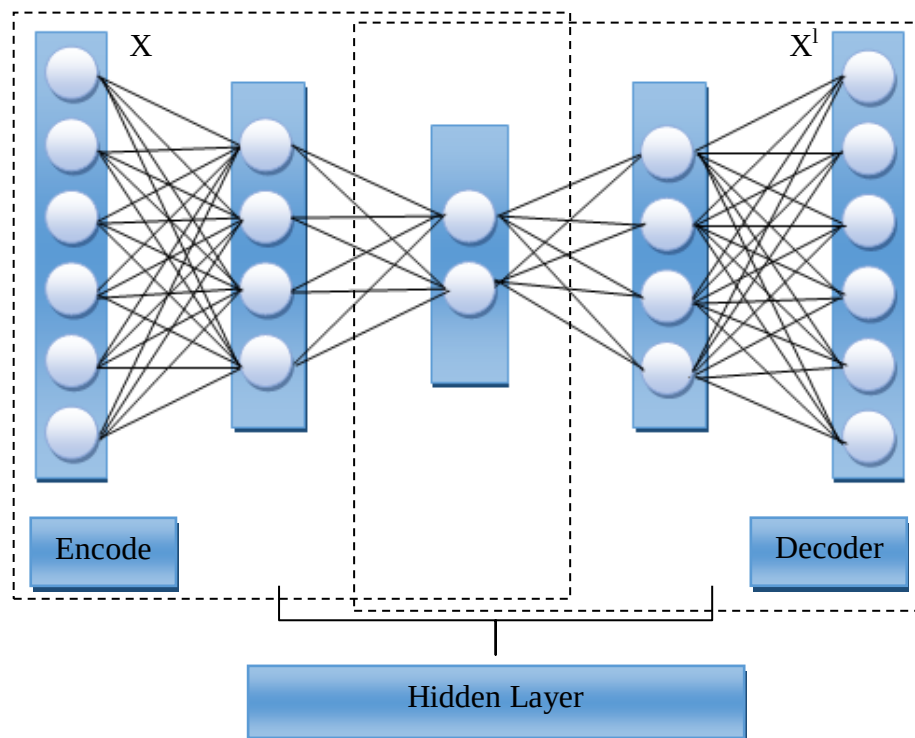


Figure 1.5. Autoencoder Method Architectures [22]

1.3.4. Sparse Coding

Sparse coding is an unsupervised algorithm that learns to represent input data succinctly using only a small number of bases [26]. Sparse coding work method, one receives samples of vectors $y \in R^n$ that have been generated as $y=A*x^*$ where $R^{n \times h}$ and $x^* \in R^h$, here number of non-zero entries in x^* to be no larger than some function of the dimension h and that A^* satisfies certain incoherence properties [23]. Sparse representation of information provides a powerful means to perform feature extraction on high-dimensional data and is of broad interest for applications in signal processing, computer vision, object recognition, and neurobiology [27]. Specially, in biometric recognition systems, sparse coding successfully was applied with high resolution results. Fingerprint biometric recognition for personal identification obtains an over complete dictionary from a set of fingerprint patches allows to represent them as a sparse linear combination of dictionary atoms [28]. In biometric classification phase, sparse coding has been successfully applied by researchers. Additionally, advantage of sparse coding has been applied on face identification system, which use sparse code as a representation of each face image, and then the Mahalanobis distance between two sparse codes is computed to determine whether or not both face images belong to the same person[29].

1.4. Recurrent Neural Network (RNN)

Recurrent neural network can use their feedback connections to store representation of recent input events in the form of activations for short-term memory, as opposed to long-term memory embodied by slowly changing weights [30]. RNN method is a very simple model compared to other neural network models as illustrated in Figure 1.6. RNN is different from feed forward networks in the way of processing input, every process in a step-by-step manner, at step t the network receives x_t as input and having knowledge about state from last step h_{t-1} it computes its output according to the formulas (1) and (2). W_{hh} , W_{xh} and W_{hy} are matrices of network parameters [31].

$$h_t = \tanh(W_{hh} \cdot h_{t-1} + W_{xh} \cdot x_t) \quad (1)$$

$$y = W_{hy} \cdot h_t \quad (2)$$

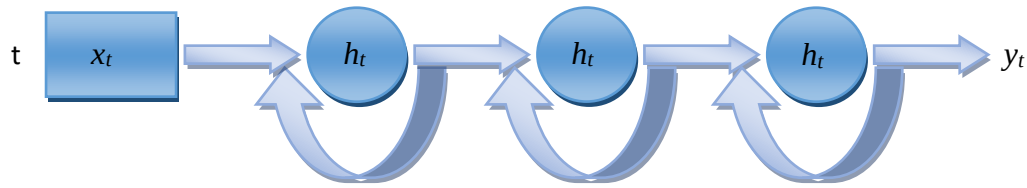


Figure 1.6. Recurrent Neural Network Architecture [30]

RNN has been applied successfully to solve machine learning problems and gained renewed results, which include sequential data. In a study, RNN successfully applied automatic Russian speech recognition[32], which able to store arbitrary long history of given word. Handwriting recognition system has successfully experimented significant features extraction from word image data and used RNN [33].

1.5. Problem Statement

In this thesis, a biometric pattern recognition system using unimodal infrared vein pattern on dorsal hand is developed. For this purpose, hand NIR (Near Infrared) images vein modalities are used. In addition a CNN deep learning method is proposed for hand dorsal vein recognition. The advantages of proposed CNN method enhanced specific model parameters to improve recognition performance. Main observations are listed below.

- Quality of raw vein image affects the pattern recognition performance. Specially, recognize specific blood vessel pattern from Illumination (here present NIR) images is the most challenge expectation.
- Different specific diameters of blood vessels may found along the dorsal hand part. Different gander and different parts of hand dorsal veins may appear differently as too thick or too thin. This difference also affects the performance of pattern recognition; it will bring more challenge in preprocessing operation.

- Traditional recognition methods have used handcrafted features to obtain features from dorsal hand vein images. Thus, to propose an efficient optimization algorithm is also a challenge in this thesis.

1.6. Aim of the Study

The aim of our study is to investigate dorsal hand vein image preprocessing to provide efficient images to vein featurize.

- To design and establish special methods to recover the quality of vein images and vein performance.
- To apply the Region of Interest (ROI) localization-based algorithm to determine the position of the ROI of hand vein images and by using CNN to predict vein from hand images.
- To compare and analyze the performances of three methods, namely Principle Component Analysis (PCA), Support Vector Machine (SVM) and CNN, in hand vein pattern recognition system.

1.7. Literature Survey

In this section, a summary of the literature on exciting biometric recognition systems on deep learning methods is given. Some key issues in biometric recognition systems are complex feature extraction in low level resolution image and exceptional recognition accuracy. Deep learning allows computational models which are composed of multiple processing layers to learn representation of data with multiple levels of abstraction [16]. In recent years, deep learning based methods successfully applied in face pattern recognition [34][35], fingerprint pattern recognition[36][37], palmprint pattern recognition[38], vein pattern recognition[38][39], eye and iris pattern recognition[40][41][42][43] and gate recognition[44][45].

Since 2000, various high recognition rates have been proposed in deep learning. Deep learning is one of the major challenges of computer vision, machine learning, and artificial intelligent (AI) in general. In the next decade, deep learning will be able to devise methods that can automatically learn good features of hierarchies from unlabeled

and labeled data using an integrated fashion [46]. There are some studies, have applied CNN on images and pattern recognition with high recognition rates.

Handwritten character recognition and text detection [47] are also challenging pattern recognitions in image vision area. In handwritten recognition, the researches attempt to activate application of CNN classify techniques on Latin handwritten digits [48], Chinese handwritten digit [49], and Arabic handwritten digit [50]. In deep learning applied to handwritten character recognition system, and explored with CNN method CNN preformed 99.28% best accuracy for recognize Latin handwritten digit dataset [51].

In biometric recognition studies, CNN has been successfully applied on face recognition [52], fingerprint recognition [53], gate recognition [45], Irish and eye recognition [41], and finger vein pattern recognition [54] problems as it has number of various advantages compared to other image and pattern recognition techniques. Face pattern is one of the most challenging applications in image recognition tasks and the application of CNN to face pattern has attracted researchers in face recognition. Particularly, CNN represented on face detection [55], face feature extraction [56], face recognition [57], and many other parts. The aim of our research is to apply LeNet-5 common architecture [58] CNN to solve non-linear effects such as illumination variances and poses for face recognition. They applied LaNet-5 CNN architecture on different two-face datasets and achieved 99.5% and 85.7% recognition accuracy rates, respectfully.

Convolutional networks and other deep models have achieved best accuracy in vision tasks such as logo recognition [59], image spam recognition [60], traffic sign recognition, food recognition [61], plant identification [62][63], automatic book detection [64], and extreme weather detection[65] etc. Among others, studies that applied CNN on vision recognition experiments have resulted in outstanding performances compare to other machine learning studies.

Some studies have focused on addressing the problems related to image and pattern recognition such as vein pattern recognition, mainly the experiments conducted on finger vein recognition dataset. There are some research propose to apply finger vein automatic recognition using advanced technology CNN and to explore the performance of the different CNN architectures.

Itqa and his colleagues [54] have introduced CNN with Graphical User Interface (GUI) as the user input approach applied finger vein recognition system. CNN architecture which was proposed in this research, was used four layers to retrain network for new input train finger vein subjects, the convolution and subsampling layers have been used, and classification has been processed through two fully connected single nodes. The final performance of finger vein recognition has given average of 96% accuracy rate by increasing train subject numbers from 1 to 10 in less than 10 second.

In a study, researchers have demonstrated four layers fused convolutional layer and subsampling layer without including input layer of CNN architecture for applying the finger vein recognition system. This proposed CNN architecture is implying 5, 13, and, 50 feature maps in the first three layers. The last layer represents the output layer which fixed with 50 neurons as perform 50 subjects target classify with 10 finger samples for each finger. An interesting performance detail in this research is the implementing the identification result with tested from 50 to 81 finger vein subjects, accuracy rates achieved 100% and 99.38% [66].

Kashihara [67] has utilized deep convolutional neural network (DCNN) as a super-resolution in order to improve the quality of downgraded images with a low level abnormal vein image. In this research, DCNN was designed with five convolutional deep layers and the size of the filter kernel has been fixed to 3×3 and exited for more suitable optimal number of feature maps. Moreover, maximum pooling layer operation has been performed by each convolutional layer. At the same time, last convolutional layer was not performed with any pooling layer, which connected with the final image output process. In the current research, the final result is consistent with the results of other techniques and DCNN is able to enhance the accuracy of complicated vein shapes by modifying filter kernels, although the estimated quality of the vein image with DCNN optimal structures and parameters.

There are other vein pattern recognitions that have been proposed based on dorsal hand vein recognition system, and received strong attention from researchers and security manufactures. CNN methods take a different approach and achieve high accuracy by applying on hand vein recognition. In another study, researchers have investigated deep learning based methods on dorsal hand vein recognition and have done a comparative

study of popular CNN architectures (i.e., AlexNet, VGG Net and GoogLeNet) for such an issue[39]. To the best of our knowledge, this study will be the first attempt that applies deep models to dorsal hand vein recognition. The evaluation is conducted on the NCUT database, and state-of-the-art accuracies are reached. Meanwhile, the experimental results have also demonstrated that the advantage of deep features to the shallow ones is to differentiate dorsal hand venous network and have confirmed the necessity of the fine-tuning phase.

CHAPTER 2

DORSAL HAND VEIN PATTERN RECOGNITION AND METHOD

2.1. Dorsal Hand Vein Recognition

Hand vein biometric recognition is defined as the use the patterns of hand blood vessels to identify an individual [68] . Hand vein images are acquired in infrared light and skin variations and dirtiness are not as apparent as in visible light [69], which is highly advanced to use in safest biometric term with spoof attacks. They are able to use the specific patterns that are very hard to duplicate, which are unique to each individual, and it is also impossible to lose or forget them since the biometric characteristics are essentially attached to the human body [68].

2.2. Experiment of Pre-processing Analysis

Since 2006, deep learning methods start to apply on Machine Learning and Artificial Intelligences areas. Deep learning strategies promise to leverage very large data sets for finding the hidden structure and also make an accurate prediction [14]. A class of machine learning techniques applied to exploit many layers of non-linear information processing for supervised or unsupervised feature extraction and transformation, and for pattern analysis and classification [15]. In this section, we discuss and summarize the empirical pre-processing analysis using deep learning approach, and CNN techniques to classify hand dorsal vein recognition, which provide the best accurate results.

The preprocessing of hand vein recognition is divided into four stages shown in Figure 2.1 the process starts with using dataset for free of charge with the permission from the head of Brain Diseases Analysis Laboratory (BDALab) at the BRNO University of Technology [76]. Images have been processed following stages below:

- i. Hand preprocessing: This is the stage to preprocess the original NIR hand dorsal vein side image, which is the procedure after system receiving the raw data [25]. First, the image banalization from RGB image, crop hand region, image rotate, and image resize process. The output is obtained using median filter [70] method to reduce image noise. Median filtering is a nonlinear method used to remove noise from images, which is widely used as it is effective at removing “salt and pepper” type noise.
- ii. Hand segmentation: This is a segmentation process done with Fuzzy C-mean algorithm, which converts a gray scale images into banalization images with black and white pixels. The segmentation techniques used is to segregate the hand finger from hand image to acquired hand dorsal part and it also filters some noise from hand dorsal site to successfully get to the process area.
- iii. Hand vein feature extraction: The region of interest algorithm is represented in this system, as ROI is referred to an area, which has represented the characteristics of a hand vein in image. Extracting the ROI from hand vein images is a main procedure of the hand vein acquisition system [71].
- iv. Classification: Three different machine learning methods were tested: Convolutional Neural Network (CNN), Principle Component Analysis (PCA) Algorithm combination with Minimum Distance Classifier (MDC), and PCA combine with Support Vector Machine (SVM) classifier. Those methods have been trained by previous stages to obtain the proposed feature.

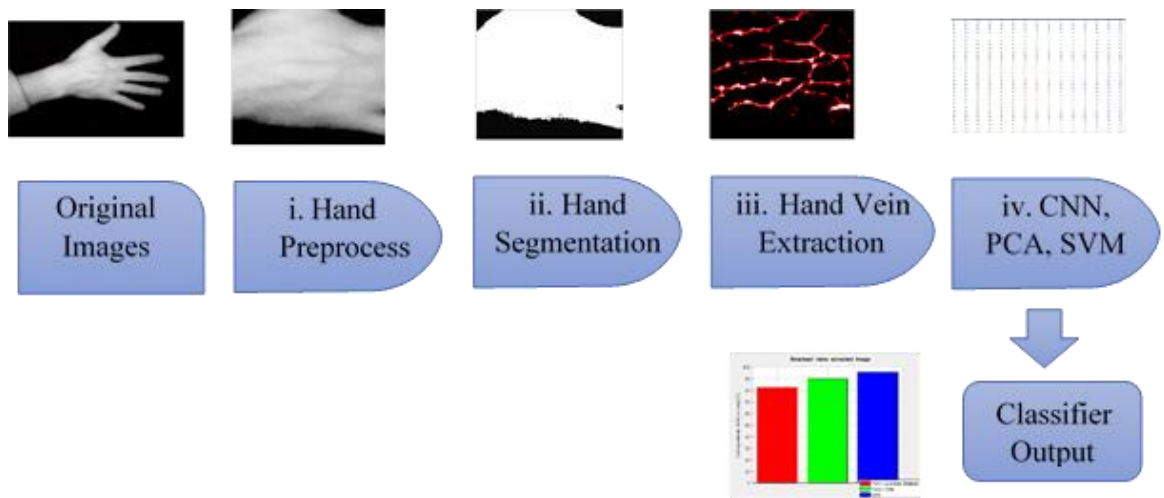


Figure 2.1. NIR Hand Vein Classification Processing

2.3. CNN Classification Model

The CNN is one of the most notable deep learning approaches, which consisted of a convolutional layer and a pooling layer [15]. It has been found highly effective and has become the most commonly used method in diverse computer vision applications. CNN is extended from artificial neural network, the convolutional layer shared many weights, and the pooling layer subsample the output of convolutional layer also reduced the data from previous layer. First, deep learning mainly applied for character recognition tasks such as reading zip codes and digit [72][73][48]. The deep learning subject used CNN method, which was the first CNN design used architecture named LeNet by Yann LeCun. The pipeline of the general CNN applied computer vision architecture is shown in Figure 2.2. Generally, a CNN consists of three main neural layers, which are convolutional layers, pooling layers, and fully connected layers. Different kinds of layers play different roles.

The CNN basic architecture design which is used in this study for hand vein recognition is shown in Figure 2.2 the main operation for vein recognition is done with five stages:

- Input images layer: CNN required a fixed input image size and we proposed input images that consist of 3 feature maps (an RGB color image) with size 640×480 .

- Convolution: For the operation, each convolution uses same kernel size 5×5 for each feature map in convolutional layer C1 and C2. The first convolutional layer C1 contains 6 feature maps and second convolutional layer C2 contains 12 feature maps.
- Pooling or subsampling: The most popular pooling layers include max-pooling and average pooling. In our system, we designed using the max-pooling layer. Pooling layers are usually applying convolutional layer, which simplify the source in the output from the convolutional layer. In the proposed system, architecture includes two subsampling layers, namely, S1 and S2. S1 layer contains 6 feature maps and a 2×2 kernel for all feature maps. For the S2 layer consists of 12 feature maps and a 2×2 kernel for all feature maps.
- Classification (Fully connected layer): This is the final layer, which is connected to all networks and also connected to every neuron from the max-pooled layer to every output neuron. Fully connected layer is used to finalize the network after feature extraction. The process was done by convolutional and pooling layers.
- Output result (non-linear or softmax): Output result will represent non-linear function of sigmoid or softmax transfer functions.

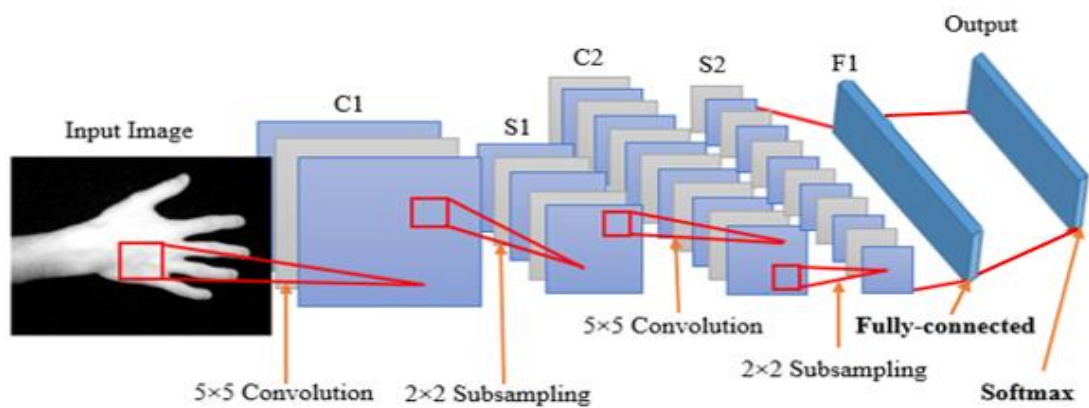


Figure 2.2. Convolutional Neural Networks LeNet-5 Architecture [46]

2.4. PCA based MDC and PCA based SVM classification Models

This study proposes a method, which is developed by combining the principle component analysis (PCA) for feature extraction and the minimum distance classifier

(MDC) for classification. PCA is a powerful tool for analyzing data, identifying patterns and expressing the data to highlight their differences [74], so PCA method is used to reduce the dimensions to provide the most distinguished features for classification.

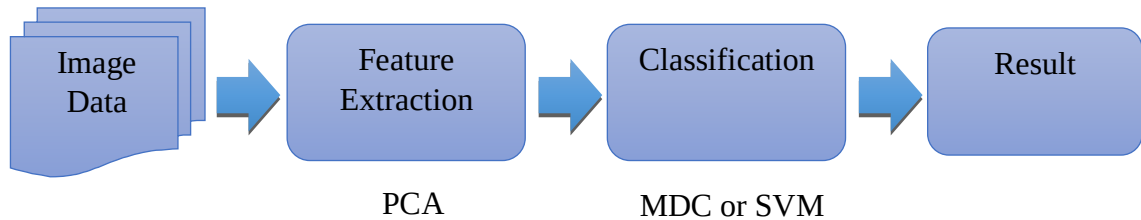


Figure 2.3. Architecture of PCA Combination with MDC or SVM Classification Models

The diagram of the hand vein recognition system using the combination of PCA with MDC method is illustrated in Figure 2.3. After the input of the data from dataset, each image is resized 50×50 to reduce the computational time, then, each image has represented in one column. In this research, we have trained 9 samples and calculated the PCA space, eigenvalues, and eigenvectors to select the principle components. After training data on the PCA spaces, each image was assessed on PCA spaces for classification. Using this method, we have performed hand dorsal vein based on Minimum Distances Classifier (MDC) or Support Vector Machine (SVM) to classify hand dorsal vein patterns from dorsal hand images. SVM is a powerful classification method used in machine learning process. It has been widely applied on pattern recognition system. The hand image data has included more classes, so we have used SVM to resolve binary classification problems in our research. The hand vein recognition is a nonlinear classification model, thus, we have applied kernel function [75] to efficiently perform the computation of input data directly in feature spaces. There are some kernel functions like linear, polynomial, RBF and sigmoid in the present study. Moreover, we have proposed the sigmoid kernel function as well as a 10 fold cross-validation was applied to calculate the misclassification rate.

CHAPTER 3

THE RESULTS OF THE PROPOSED ALGORITHM

In this thesis, CNN method is applied for the recognition of hand dorsal vein patterns. From the CNN design architecture, every convolutional layer is represented by the size and the number of the maps, and kernel size parameters. The parameter values are compared based on the CNN classification accuracy and time. The experimental results are conducted in MATLAB R2015a programming environment. Furthermore, LeNet-5 network architecture is used for implementing CNN on hand dorsal vein recognition.

3.1. Dataset

In order to design reliable hand vein recognition system, we need to use a high resolution thermal hand image database. The database in this study has been obtained from the Brain Diseases Analysis Laboratory (BDALab) at the BRNO University of Technology [76].

Hand dorsal vein images have been collected in different environment using 10 different hands for each participant. There are 100 participants altogether, with 10 samples from each participant. Hence, there are a total of 1000 samples. In this thesis, we used only 60 participants' data from the dataset. Figure 3.1 shows a sample hand dorsal image from (BDALab) dataset.

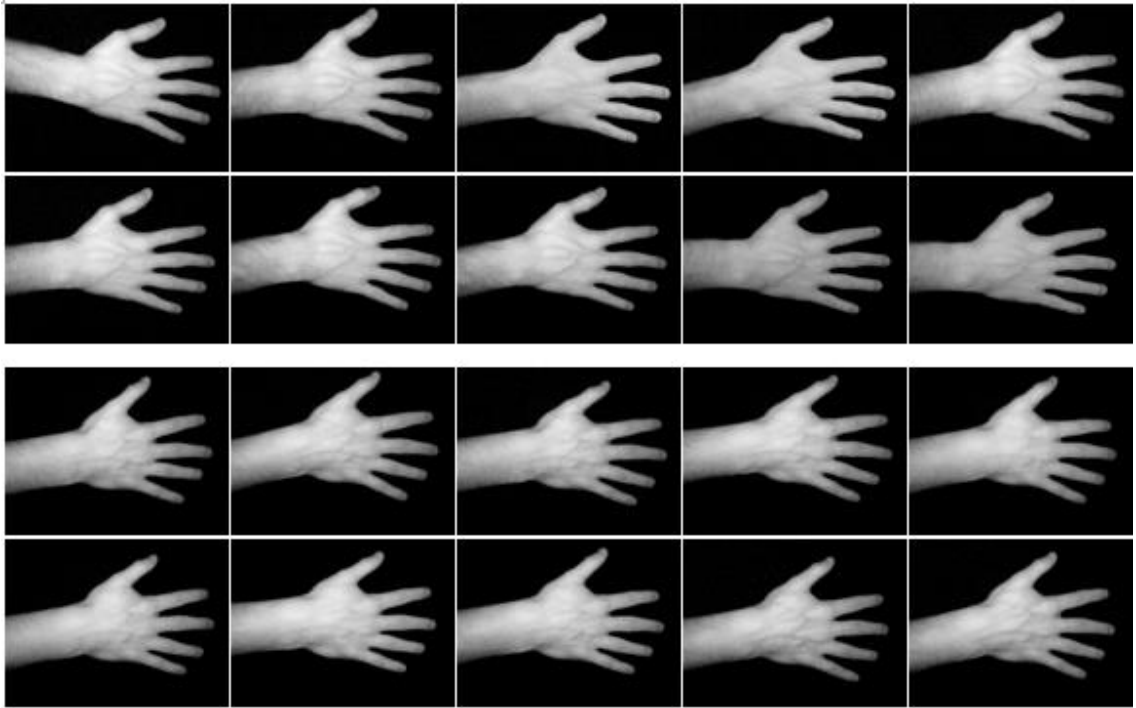


Figure 3.1. A Sample of Hand Dorsal Image from the BDLab Dataset [76]

3.2. ROI Feature Extraction and CNN Classification

To analyze hand dorsal vein recognition system, we need to first perform preprocessing step, then allocate the region of interest feature extraction technique, and finally apply the CNN as a classification method to recognize the vein pattern on samples. Figure 3.2 depicts the main GUI design which includes all parts of the proposed method. The GUI contains several information such as the image preprocessing, FCM image segmentation, image feature extraction, and image classification.

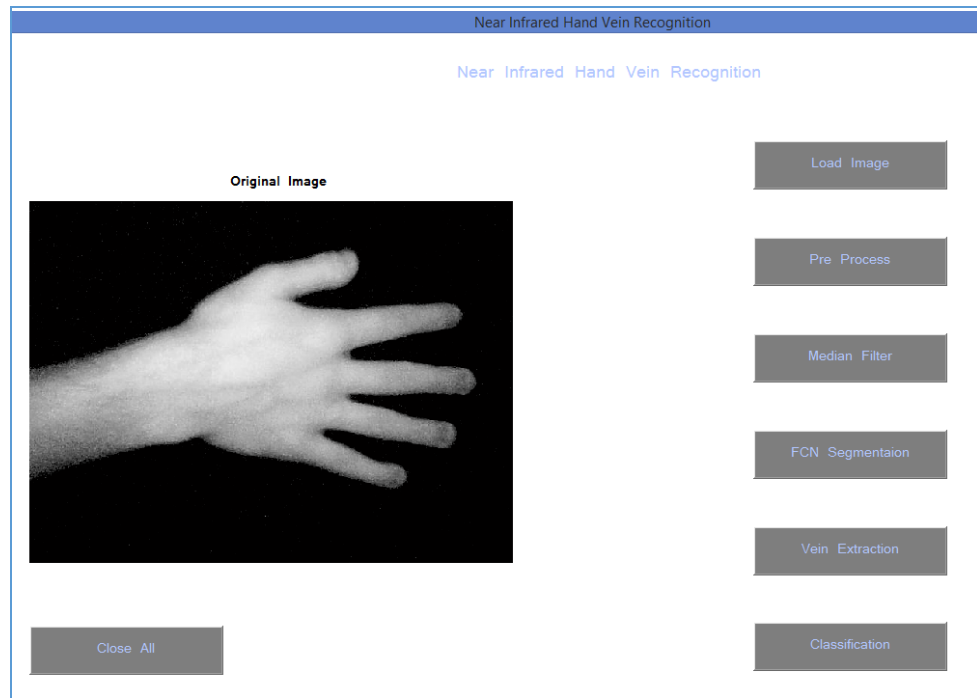


Figure 3.2. GUI of the Developed Application

Hand image samples will be selected from the dataset through “load image” to operate preprocessing stage. The original image size in the database is 640×480 and after images are cropped, the region hand dorsal image size reduces to 250×250 . Figure 3.3 illustrates image preprocess stage results.

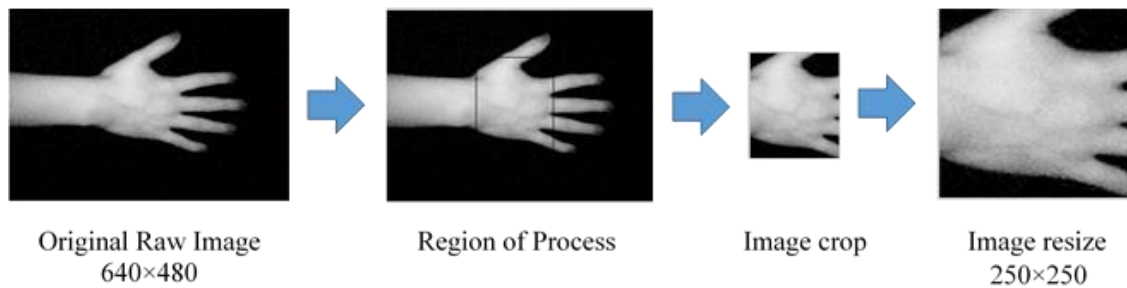


Figure 3.3. Result of Image Preprocessing for Hand Vein

When the images have been preprocessed, next step is to apply median filter to reduce image noise, and the image segmentation which is done by Fuzzy C-means clustering method. Hand dorsal part segmentation is key to enable integrated hand dorsal vein region part for vein recognition. This segmentation task will pose challenges by the fact that the hand dorsal vein region may overlaps with the cropped section. The Fuzzy C-means clustering method brings better segmentation results and improves feature

extraction accuracy. Figure 3.4 demonstrates the procedure of segmentation with two cluster.

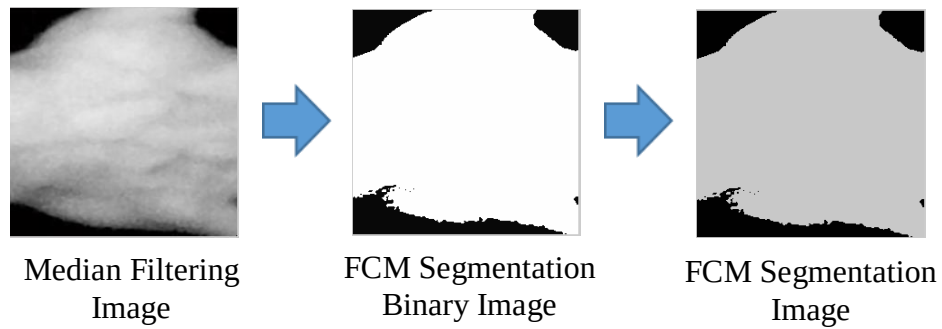


Figure 3.4. Result of Fuzzy C-means Clustering Methods Represent on Hand Dorsal Site

Hand dorsal vein feature extraction model follows the hand vein template [68], which will be used for classification stage. In this thesis, we used the convolutional neural network for classification step. Feature extraction stage has been developed using the Region of Interest (ROI) method. In this thesis, we have used crow search algorithm [77] to select vein extraction area, and extracted vein from the hand dorsal site with maximum curvatures method [78], which try to extract the center lines of the veins on the dorsal hand. Figure 3.5 shows results from vein features extraction of the hand dorsal images.

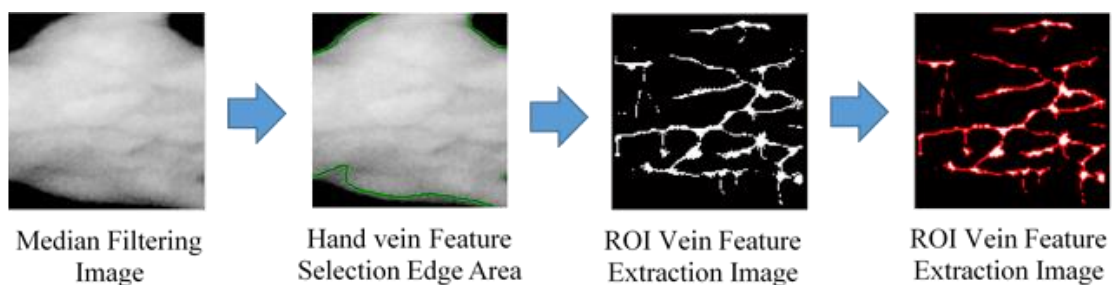


Figure 3.5. Result of Region of Interest (ROI) Feature Extraction Methods Represent on Hand Dorsal Site

After the preprocessing steps, CNN classification method is shown in Figure 3.6 is conducted. In here CNN will recognize veins from dorsal hand.

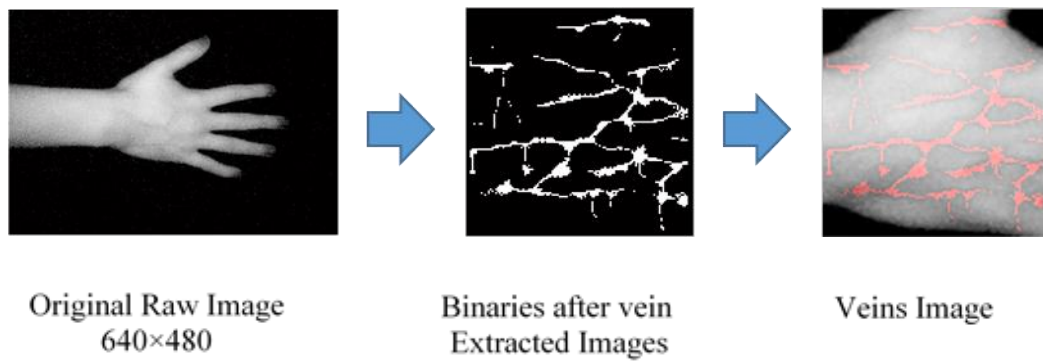


Figure 3.6. CNN Classification Hand Vein on Hand Dorsal Site

CHAPTER 4

CONCLUSION, DISCUSSION AND FUTURE WORK

In this chapter, we compare and discuss the result of three different models in the same dataset. In this thesis, algorithms of CNN, PSA and SVM applied on NIR Dorsal Hand vein pattern recognition. Results show that CNN classification provided better result in term of accuracy compare to PCA with MDC and PCA with SVM.

4.1. CNN Classification

Table 4.1 compares the accuracy and the time of CNN method in terms of epoch number. It clearly shows that epoch 4 outperforms in both accuracy and time measures; in our CNN design and the accuracy for epochs 4 is 94.5%.

Table 4.1. The Accuracy Result of Epochs Changes 1-10

Epochs	Time (second)	Accuracy (%)
1	17.6245	91.5
2	17.2147	90.5
3	16.8478	93.5
4	16.8844	94.5
5	17.0674	92.5
6	16.9596	91.5
7	16.9035	90.5
8	17.0705	90.5
9	17.0997	92.5
10	17.3611	90.5

4.2. PCA based MDC Classification

PCA based MDC classification presented previous research as refer to as mentioned chapter 2 and chapter 3. This model developed by combining PCA for dorsal hand vein feature extraction and MDC for classification. Experiment done on dataset (BDALab) and recognition vein rate achieved to 84%. The results of the PCA based on MDC and PCA based on SVM methods are given in Table 4.2.

Table 4.2. The Result of PCA, MDC and SVM Methods

Recognition Methods	Accuracy (%)
PCA+MDC	84.0
PCA+SVM	90.0

4.3. PCA based SVM Classification

PCA based SVM classification is performed on the same dataset. PCA is used to extract key vein features from dorsal hand vein images and SVM is used use for vein classification. We illustrate the SVMs on dataset, which consider of 60 individuals with 600 images, and achieved high performance on vein details. We proposed the sigmoid kernel function in our PCA based SVM model to classified vein performance. In the end, the classification accuracy in classifying vein pattern from the dorsal hand images is 90% shows in Table 4.2.

Figure 4.1 shows the performance of vein pattern recognition of CNN with PCA and SVM. It shows that the experimental lowest accuracy results has been acquired by PCA method applied with MDC classification method which is 84% and best accuracy is obtained by CNN outperformed the PCA methods with the accuracy 94.5%. The second highest accuracy by PCA based SVM classification method obtained with 90%.

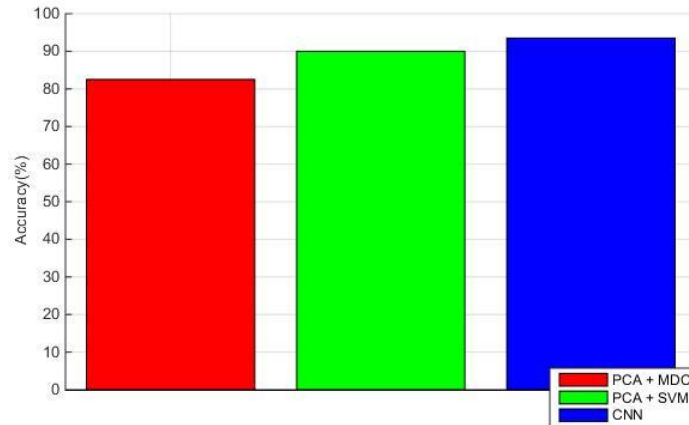


Figure 4.1. Comparison of Performances of Hand Dorsal Vein Recognition of Methods

4.4. Conclusion

This study focuses on developing and presenting deep learning method based on CNN and its application on the Dorsal Hand Vein Pattern Recognition.

In this study, there have been reported a successful application of deep learning to vein recognition in different temperature conduction, especially to hand dorsal vein pattern recognition. We proposed to use CNN for dorsal hand vein recognition. Also we proposed dorsal hand vein recognition based on PCA, MDC and SVM methods. PCA is used for vein feature extraction and dimensional reduction. Experimental results show that the best accuracy is obtained using a standard parameters present in the CNN model. In the proposed method, it is not necessary to handcraft a specific feature extraction method.

In the future work, we plan to focus on hand dorsal vein recognition for person identification and robot injection system.

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CURRICULUM VITAE

Name and surname: SAINAIMUJIANG KUERBAN

Nationality: CHINA

Birth date and place: 27-12-1976 / China- Urumchi

Marital status: Single

Cell phone: 05454266827

E-mail: sanamjan328@gmail.com

Correspondence Address: Bahcleievler Mah. AFM Newstar, Cblok, NO: 2, Talas,
Kayseri, Turkey.

EDUCATION

Degree	Institution	Date of graduation
MSc	Erciyes University	2018
Undergraduate	Chang'an University	1999
High school	Urumchi High School	1993

FOREIGN LANGUAGE

Uyghur

Turkish

English

Chinese

Malay
